DEGREE PROJECT

Aware of the Feed

Algorithmic Media Content Awareness and Negative Social Comparison on Social Media

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Foreword

I would like to express my gratitude to everyone who contributed to this study. I am grateful to the participants who took the time to complete the survey and make this research possible.

I would also like to thank my supervisor for guidance and feedback throughout the project.

Finally, I would like to thank my partner for support and encouragement during the writing process, and my son, who spent many hours sleeping peacefully while unknowingly keeping me company as this thesis was being completed.

Katrin Isaksson Kyrk

Abstract

The purpose of this study was to investigate the relationship between algorithmic media content awareness and negative social comparison among upper secondary school students. The study draws on social comparison theory, social media literacy research, and research on algorithmic media content awareness. While previous research has shown that social media may intensify negative social comparison through features such as photo-editing, filters, and visible engagement metrics, little is known about how awareness of algorithmic content curation relates to these processes.

The study adopted a quantitative cross-sectional survey design. Data were collected through an online questionnaire administered to upper secondary school students in Sweden (N = 63). Algorithmic media content awareness was measured using an adapted version of the AMCA scale developed by Zarouali et al. (2021), while negative social comparison was measured using adapted items based on De Vries and Kühne (2015). Pearson correlation analysis was used to analyze the data.

The results showed a very weak negative relationship between the variables, but the relationship was not statistically significant. Thus, the hypothesis was not supported. However, exploratory analyses suggested that the subdimension human-algorithm interplay may be relevant in relation to certain forms of negative social comparison. The study contributes to the literature by examining a previously unexplored relationship and highlights the need for further research within this area.

Keywords: algorithmic media content awareness, negative social comparison, social media literacy, social comparison theory, social media, adolescents.

Sammanfattning

Syftet med denna studie var att undersöka sambandet mellan *algorithmic media content awareness* (medvetenhet om algoritmisk kurering av medieinnehåll) och negativ social jämförelse bland gymnasieelever. Studien utgår från *social comparison theory* (social jämförelseteori), forskning om *social media literacy* (medie- och informationskunnighet i sociala medier) samt forskning om *algorithmic media content awareness*. Tidigare forskning har visat att sociala medier kan förstärka negativ social jämförelse genom funktioner såsom bildredigering, filter, och synliga mått på engagemang. Däremot är kunskapen begränsad om hur medvetenhet om algoritmisk kurering av medieinnehåll relaterar till dessa processer.

Studien genomfördes med en kvantitativ tvärsnittsdesign. Data samlades in genom en webbenkät som besvarades av gymnasieelever i Sverige (N = 63). *Algorithmic media content awareness* mättes med en anpassad version av AMCA-skalan utvecklad av Zarouali et al. (2021), medan negativ social jämförelse mättes med anpassade frågor baserade på De Vries och Kühne (2015). Data analyserades med hjälp av Pearsons korrelationsanalys.

Resultaten visade ett mycket svagt negativt samband mellan variablerna, men sambandet var inte statistiskt signifikant. Hypotesen fick därmed inget stöd. De explorativa analyserna indikerade dock att subdimensionen *human-algorithm interplay* kan vara relevant i relation till vissa former av negativ social jämförelse. Studien bidrar till litteraturen genom att undersöka ett tidigare outforskat samband och belyser behovet av fortsatt forskning inom området.

Nyckelord: *algorithmic media content awareness*, negativ social jämförelse, *social media literacy*, social jämförelseteori, sociala medier, gymnasieelever.

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1. Introduction

Recently, the topic of young people on social media has received a lot of media and political attention. From Australia banning people under the age of 16 from using social media (Livingstone, 2025), to Sweden enforcing a national ban on cell phone usage across all elementary schools in the country (Hogenäs, 2025), the concern regarding potential harm from social media usage among young people is highly prevalent in today's society. Thus, the topic of social media usage has been widely studied in the literature on information systems (e.g. Krasnova et al., 2015; Li, 2017; Meier & Schäfer, 2018; Meythaler et al., 2025; Wenninger et al., 2021). Researchers are concerned with how platform design aspects might influence the well-being of users (e.g. Meythaler et al., 2025; Wenninger et al., 2021), as well as with how potential negative effects might be mitigated through various intervention strategies (e.g. Paxton et al., 2022; Schaben et al., 2026).

One aspect related to the functionality of social media platforms that has received attention recently both in the media (Bergendorff, 2025) and in the academic literature (e.g. Issar, 2024; Su et al., 2021) concerns the growing prevalence of algorithmic content curation. In court documents from the summer of 2025, it was confirmed that as little as 7% of the time users spend on Instagram involves engaging with content from accounts they actively follow. Instead, the content that users see and interact with on social media platforms like Instagram and Facebook are mostly recommended by AI-powered algorithms (Federal Trade Commission v. Meta Platforms, Inc., 2025). Algorithmic content curation is mainly driven by the desire to increase user engagement (Carrasco-Farré et al., 2025), and researchers have been able to show a difference in brain region activation when comparing exposure to algorithmically curated video clips to non-personalized video clips (Su et al., 2021). Similarly, the Public Health Agency of Sweden (2025a) has raised concerns about how many digital platforms are intentionally designed to increase the amount of time users spend online, partly through algorithmic recommender systems. In the literature, having an understanding of the ways in which algorithms are used for content curation on media platforms is referred to as algorithmic media content awareness (Zarouali et al., 2021).

While the research on social comparison processes on social media is extensive (e.g. Krasnova et al., 2015; Li, 2017; Meier & Schäfer, 2018; Meythaler et al., 2025; Wenninger et al., 2021), and algorithmic media content awareness has been studied in areas such as TikTok-addiction (Wang & Guo, 2023), a search of the relevant literature did not identify any studies examining the relationship between social comparison and algorithmic media content awareness. Thus, the present study aims to address this gap in the literature. Moreover, besides potential theoretical contributions to the research field, having an understanding of the ways in which algorithmic awareness might influence social comparison processes on social media could have practical implications for the development of social media literacy interventions. Currently, algorithmic

awareness is not a standard feature of such interventions (e.g. Schaben et al., 2026), which may constitute a blind spot in media literacy programs given the extensive prevalence of algorithmic content curation in today's media landscape. Particularly for adolescents, this knowledge is especially useful given how adolescence is often characterized as a period of self-discovery and identity formation (Maree, 2021). In addition, the Public Health Agency of Sweden (2025b) notes that adolescents generally have a more difficult time regulating their screen time and social media use, due to their long-term consequence thinking not being fully developed. Hence, young people may be especially vulnerable to the negative effects of social comparisons on social media.

1.1 Research Purpose

Despite growing societal concern regarding algorithmically curated social media environments, little is known about whether algorithmic media content awareness is related to negative social comparison. More knowledge is thus needed regarding whether algorithmic awareness may be relevant in efforts to reduce harmful social comparison processes among young social media users. Such knowledge may be useful for informing the development of social media literacy interventions and public recommendations regarding young people's social media use. Therefore, the purpose of the present study is to investigate the relationship between algorithmic media content awareness and negative social comparison among upper secondary school students.

1.2 Delimitations

This study is delimited to examining the relationship between algorithmic media content awareness and negative social comparison among upper secondary school students. The empirical investigation focuses on students enrolled in vocational programs at an upper secondary school in a smaller town in central Sweden. Furthermore, the study focuses specifically on negative social comparison and does not examine other potential outcomes associated with algorithmic media content awareness. Finally, the study adopts a hypothesis-testing approach and is limited to examining whether a statistical relationship exists between the variables. It does not investigate causal mechanisms or participants' experiences in depth. Together, these delimitations help ensure a clear and focused analysis of the relationship between the variables.

2. Theoretical background and hypotheses development

The present study draws on three theoretical perspectives to examine social comparison processes on algorithmically curated social media platforms. First, social comparison theory provides the overarching framework for understanding how individuals evaluate themselves in relation to others. Second, research on social media literacy highlights how critical engagement with content posted on social media can mitigate the potentially negative effects of such comparisons. Third, the concept of algorithmic media content awareness extends this emphasis on critical engagement by focusing on users' understanding of algorithmic curation of content exposure on social media.

Taken together, these three perspectives provide a framework for explaining how algorithmically curated digital platforms may influence social comparison processes, and how awareness of these mechanisms may buffer against potential negative effects of such comparisons. Based on this theoretical foundation, a hypothesis regarding the relationship between algorithmic media content awareness and negative social comparison is developed and subsequently tested.

2.1 Social Comparison Theory

2.1.1 Core assumptions and processes

Social comparison theory was originally articulated by Festinger (1954) and has since been widely applied across disciplines. According to the theory, individuals evaluate their abilities, status, and self-worth through comparison with others. Upward comparison entails comparison with someone who is regarded as superior to oneself (Meythaler et al., 2025), whereas oneself is considered superior to the other in instances of downward comparison (Krasnova et al., 2015). There are several factors that influence the relevancy and impact of social comparisons (Festinger, 1954). The similarity between oneself and the target of comparison matters, since we can gain a more accurate estimation of our own standing through comparison with someone who is more similar. Furthermore, the effects of social comparisons are intensified when it relates to an aspect of ourselves that we consider to be important (Wenninger et al., 2021). Thus, upward comparisons to similar others in self-relevant domains can cause feelings of envy to arise, which in turn have been associated with detrimental effects on subjective well-being and quality of life (Krasnova et al., 2015).

Given this background, it is not surprising that upward social comparisons are often associated with negative effects on well being (Krasnova et al., 2015; Meythaler et al., 2025; Wenninger et al., 2021). However, some scholars note that upward social comparisons can also have beneficial effects, and that it is important to distinguish assimilative comparisons from contrastive comparisons (Meier & Schäfer, 2018). As mentioned by Turel (2021), engaging in upward comparisons also entails a process of subconscious goal-setting to bring one closer to the desired

state. This is the focus in assimilative comparisons, where one aspires to become more similar to the person to whom one is comparing oneself, which in turn leads the way to inspiration and positive action (Meier & Schäfer, 2018).

Contrastive upward comparisons on the other hand, entails bringing the differences between oneself and the desired state of the other to the forefront. In such comparisons, attention is directed not only toward the gap between oneself and the comparison target, but also toward the factors perceived to maintain these differences. This may lead individuals to feel that they are unable to achieve the same desirable qualities or success as the comparison target, which in turn makes contrastive upward comparisons more closely associated with negative outcomes (Meier & Schäfer, 2018). For example, a student viewing social media posts from peers or influencers who appear more attractive, popular or successful may focus on perceived disadvantages in their own life that make such achievements seem difficult or unattainable, rather than feeling inspired.

One key concept related to the purposes of the present study is "negative social comparison". De Vries and Kühne (2015) define this concept as "feeling like others are doing better than oneself" (s. 1). In relation to social comparison theory, negative social comparison can be understood as a more negative form of upward social comparison (such as contrastive upward comparisons), where the comparison leads to feelings of inferiority and/or inadequacy.

At the same time, it is important to note that upward social comparison is a broader concept. As discussed previously, upward comparisons do not necessarily lead to negative outcomes, since some upward comparisons may also be experienced as motivating or inspiring (Meier & Schäfer, 2018). However, the present study focuses primarily on the more negative and contrastive forms of upward comparison that are associated with poorer self-perceptions and reduced well-being.

Because different researchers use somewhat different terminology, both the terms "upward social comparison" and "negative social comparison" will be used throughout this paper depending on the terminology used in the cited source material.

2.1.2 Social comparison in IS research

In IS research, social comparison theory is frequently used as a framework to discuss the effects of social networking sites (SNS) on well-being (Krasnova et al., 2015; Li, 2017; Meier & Schäfer, 2018; Meythaler et al., 2025; Wenninger et al., 2021). There are several socio-technical aspects pertaining to the design of SNS and other forms of social media that can cause them to amplify the effects of social comparison processes, as compared to an offline-setting. First, researchers have identified the presence of a positivity bias on SNS, meaning that users tend to curate their posts intentionally to create a more aspirational picture of themselves and their lives (Wenninger et al., 2021). The availability of photo-editing and filters further enables the poster to alter the image being portrayed to their audience (Meythaler et al., 2025). Thus, it is not

surprising that content on SNS often includes travel, food, and joyful experiences (Wenninger et al., 2021). The highly visual aspect of many SNS also lend themselves particularly well to social comparisons regarding appearances, fitness and aesthetics. Researchers have observed that exposure to attractive people on SNS are associated with feelings of inferiority and envy, often explained through the mechanisms of upward social comparisons (Rosenthal-von der Pütten et al., 2019).

Another recently discussed socio-technical aspect of SNS design that can amplify the effects of social comparisons is the presence of numerical engagement metrics such as likes, comments and view counts that essentially function as status indicators. Viewing a higher-metric based digital status profile on Instagram, characterized by a high number of followers, likes and comments, was shown to increase feelings of envy towards the owner of the account through its influence on perceived socioeconomic and sociometric status (Meythaler et al., 2025). Likewise, upward social comparison in terms of likes received on a selfie (meaning that another person's selfie received more likes than one's own) has been shown to increase feelings of sadness, inferiority, jealousy and resentment (Rosenthal-von der Pütten et al., 2019). Thus, as a standard feature applied across many social media platforms, engagement metrics can in and of themselves constitute a salient domain for social comparisons.

2.2 Social media literacy

Social media literacy has been conceptualized as the ability to engage critically with the content that one encounters on social media platforms. It entails reflecting upon the motives behind why images are being displayed a certain way and what is implicitly conveyed, how the portrayals may differ from actual reality, and who might benefit from the viewer feeling a certain way about what is shown to them (Paxton et al., 2022). Thus, social media literacy has been proposed as a protective factor against negative effects associated with comparisons made on social media. For example, Paxton et al. (2022) summarizes evidence from various studies that show how knowledge regarding the fact that images on social media are frequently altered in various ways, and that the way people appear to look in photos may not accurately reflect how they look in real life, ameliorated negative effects on body image when subjects were exposed to idealized appearance portrayals.

The protective effects on social comparisons associated with social media literacy can be explained through different mechanisms. One of the most central ones, as discussed in Paxton et al. (2022), is through disrupting automatic internalization of ideals portrayed on social media and rendering any upward comparisons less influential. By recognizing the altered nature of social media portrayals, the standards set by upward comparison targets can be questioned, and any unflattering comparisons can be attributed to behind-the-scenes image-editing capabilities and ulterior motives (such as economic incentives) rather than personal shortcomings. Moreover,

being able to recognize the fact that viewing certain forms of media content has a negative effect on one's body image makes it possible to actively limit one's exposure to such content in favour of content more aligned with one's values (Paxton, 2022).

The empirical evidence supporting positive effects of social media literacy interventions is however limited, and studies show mixed results (e.g. Paxton et al., 2022; Schaben et al., 2026). Gender has been mentioned as a potential modifying variable, and it has been speculated that boys and men differ from girls and women in their engagement with social media and tendency to internalize appearance ideals (Paxton et al., 2022; Schaben et al., 2026). Also, social media literacy interventions vary in their designs, and may focus on different aspects of social media consumption and critical thinking skills. In the intervention described by Schaben et al. (2026), modules included information about body image, common editing tools, internalization processes, reframing techniques, and strategies for boundary setting, among other things.

While many social media literacy interventions emphasize the manipulated or idealized nature of social media portrayals (Paxton et al., 2022; Schaben et al., 2026), comparatively little attention has been given to the role of algorithmic content curation in shaping users' exposure to various types of content. Awareness of these algorithmic processes may therefore represent an important, yet underexplored, aspect of social media literacy. The concept of algorithmic awareness, which will be discussed in the following section, addresses this aspect more directly.

2.3 Algorithmic Media Content Awareness

The concept of algorithmic awareness has been researched across a variety of contexts. For example, previous studies have examined awareness of algorithmic decision-making in areas such as employment, politics, and policing, where algorithms are increasingly used to influence or automate decisions that affect individuals and society (Gran et al., 2021). Research has also explored algorithmic awareness in relation to privacy concerns (Shin et al., 2022) and social media use (e.g. Issar, 2024; Wang & Guo, 2023). While precise definitions vary slightly, scholars generally agree that algorithmic awareness entails having some level of understanding of the presence and functioning of algorithmic systems in the digital realm (Shin et al., 2022; Wang & Guo, 2023; Zarouali et al., 2021).

Within the context of social media, the broader concept of algorithmic awareness has been further specified as algorithmic media content awareness. This concept refers to users' understanding of how algorithms are used to recommend and selectively present content on online media platforms. Zarouali et al. (2021) have identified four distinct subdimensions that together constitute their conceptualization of algorithmic media content awareness: 1) filtering, 2) automated decision-making, 3) human-algorithm interplay, and 4) ethical considerations. These four dimensions are elaborated on in the following paragraphs.

2.3.1 Filtering

Algorithmic awareness, according to Zarouali et al. (2021), starts with recognizing that algorithms filter content on social media platforms. This means that the algorithm decides what is shown to a particular user, and what is not. As stated by Carrasco-Farré et al. (2025), one of the key incentives behind this filtering process is generating greater advertising revenue, by filtering content in a way that aims to maximize user engagement and time spent on the platform. The filtering process and curation of content exposure, based on the presumed interests of each individual user, also increases the efficiency of targeted advertisements, which is another way the social media companies can increase their monetary gains (Carrasco-Farré et al., 2025). While some users may appreciate the filtering process, as it provides easy and efficient access to content they will presumably be interested in (Shin et al., 2022), the end-goal of maximizing revenue streams may come at a cost for the long-term wellbeing for individuals and society as a whole (Carrasco-Farré et al., 2025; Newell & Marabelli, 2015). Thus, being algorithmically aware means recognizing the fact that the content one is exposed to on digital platforms is algorithmically filtered and curated, and that the filtering process is neither neutral nor random. As an extension of this awareness, it also means recognizing that the content that other people are exposed to can be vastly different from your own "feed".

2.3.2 Automated decision-making

The second dimension proposed by Zarouali et al. (2021) includes automated-decision making. This dimension highlights the fact that the algorithm automatically decides what is shown to the user and what is filtered out, often without direct human involvement or oversight. Newell and Marabelli (2015) raises a couple of issues associated with this automated decision-making process. In their paper outlining potential long-term societal effects of "datification", they use the term "little data" to denote data collection that is targeted at one individual specifically, and which is subsequently used by the algorithm to make automated decisions for that one individual. The capturing of little data is of course done by collecting and analyzing "big data" from a multitude of different sources, but rather than make predictions on a group level, little data is used to make predictions on an individual level. Such predictions can include a person's interests, motivations, spending habits, and other aspects of interests for organizations and various third party actors that can be used for monetary gain. One key issue related to automated decision making highlighted by Newell and Marabelli (2015) is the lack of explainability, and thus the legitimacy, associated with such a decision-making process. If automated decision making leads to decisions that are disadvantageous, unjust, or even actively causes harm to an individual, how will the reasoning be justified? As stated by Newell and Marabelli (2015), the complexities behind the algorithmic computations of little data are so vast and intricate that not even the makers of the algorithms can justifiably explain how decisions are being made for one person, thus the predictions made are a classic case of black-box processing.

2.3.3 Human-algorithm interplay

The topic of little data ties into the third dimension of algorithmic media content awareness, which is human-algorithm interplay. This third dimension proposed by Zarouali et al. (2021) explains that the filtering process is influenced by the behavior of the user. What the algorithm decides to show one person on one platform is not predetermined at the development stage, but actively tailored to each individual based on their behaviour on the platform and other sources of little data. Newell and Marabelli (2015) uses advertising as an example to illustrate how our online activity is taken into account when the algorithm decides on what content is suitable to display. As they mention, if we have previously shown interest in a certain product by clicking on an advertisement we are much more likely to be shown even more content related to this particular product. Carrasco-Farré et al. (2025) further highlights the human-algorithm interplay by applying a socio-technical perspective and stating that the algorithmically driven content curation subsequently influences user behaviour, furthering the feedback-loop of user activity affecting content filtering which then influences user behaviour, and so on. Being aware of this interplay means recognizing that one's own actions, such as clicks, likes, or viewing time, actively contribute to shaping future content exposure.

2.3.4 Ethical considerations

The fourth dimension proposed by Zarouali et al. (2021) is termed "ethical considerations" and refers to users' awareness of ethical issues related to algorithmically curated content. In line with prior research, this dimension encompasses issues related to privacy intrusion, lack of transparency, and algorithmic bias, which are discussed in the following sections.

2.3.4.1 Privacy intrusion

According to Zarouali et al. (2021), privacy intrusion is an ethical concern associated with the use of algorithms in digital spaces due to the accompanying need for data gathering and storing. As Newell and Marabelli (2015) note in their discussion of little data and the future of algorithmic decision-making, "businesses will know almost everything about consumers" (p. 11). Furthermore, the sources from which little data can be collected are multifold (Newell & Marabelli, 2015), and the volumes of user data required to enable personalized recommendations are substantial (Shin et al., 2022). At the same time, it is not always obvious to the user how the algorithm tracks their activities and what data is being gathered when spending time on various platforms. Another key point highlighted by Zarouali et al. (2021) is the use of data modeling to infer undisclosed personal information about the user. As Bozdag (2013) explains, a typical recommendation system uses contextual clues to predict user traits, not explicitly queried information, and the assumptions and inferences made by the algorithm could relate to private topics that the user would not want to disclose. Thus, according to Shin et al. (2022), algorithm awareness entails considering these privacy related issues and making active choices regarding

what information one chooses to disclose, and reflecting upon how one's private data may be used for algorithmic decision-making.

2.3.4.2 Lack of transparency

Transparency constitutes another central ethical concern in discussions of algorithmically curated media platforms. A recurring critique in the literature is that the content filtering is rarely made explicit to users, and neither are the data gathering and algorithmic decision-making processes (Bozdag, 2013; Shin et al., 2022). As mentioned previously, the "black-box" nature of algorithmic decision making risks undermining the legitimacy of algorithmic outputs (Newell & Marabelli, 2015). The concept of explainability is closely connected to transparency, and according to Shin (2021) explainability is a central feature in terms of affecting perceived transparency and trust in algorithmic outputs. As argued by Shin (2021), recommendations made by algorithms are more readily accepted by users when one is provided with explanations that are perceived to be fair, accurate and reasonable. Greater insight in terms of what data are being used for algorithmic calculations, as well as regarding how different sources of input data are weighed against each other, can encourage users to engage with algorithmic outputs critically rather than blindly trust or indiscriminately distrust algorithmic decision-making. Thus, reflecting upon transparency issues is a key factor in algorithmic awareness.

2.3.4.3 Algorithmic bias

Algorithmic bias constitutes the third ethical concern highlighted by Zarouali et al. (2021). There are many definitions of algorithmic bias available in the literature (Kordzadeh & Ghasemaghaei, 2022). Some focus on how pre-existing societal biases and systematic discriminations are reinforced by algorithmic outputs, such as when an algorithm trained on a biased dataset uses these biases as information for future decision-making. One classic example to highlight this phenomenon is when an algorithm used for hiring purposes is trained on a dataset where the majority of successful candidates are men due to pre-existing societal structures, and thus includes gender as a key factor in its decision-making process, favoring men over women for future hiring prospects. Another way to conceptualize algorithmic bias entails a non-neutral filtering process, where some categories of content are prioritized over others in a way that causes systematic suppression or amplification of certain types of content (Carrasco-Farré et al., 2025).

As mentioned previously, the driving forces behind recommender systems in social media systematically prioritizing certain content types over others are often tied to economic incentives (Carrasco-Farré et al., 2025). The strive towards maximizing user engagement and time spent on the platform, combined with exposing users to advertisements that are tailored to their specific presumed interests and needs, can result in a systematic over-exposure to content that triggers strong emotional reactions in viewers. While the formation of filter bubbles and echo chambers is one problematic aspect of a biased recommender system, as they reduce exposure to diverse

perspectives and reinforce pre-existing beliefs (Bozdag, 2013), bias may also operate through mechanisms beyond likeness and similarity. In their analysis of the recommender system utilized by Youtube, Carrasco-Farré et al. (2025) found that while the recommender system did prioritize videos aligned with users' inferred political views, it concurrently recommended videos that expressed more extreme viewpoints associated with the opposite side of the political spectrum, while suppressing more moderate voices from the "opposing side". The authors thus suggest that the algorithm strategically prioritizes content that is predicted to create strong emotional reactions in the user, regardless of whether this could lead to more intergroup hostility and cause negative long-term effects for both individuals and society as a whole.

Thus, awareness of algorithmic bias involves recognising that content curation is shaped by training data, economic incentives and design choices, and that algorithmic outputs may systematically favour certain perspectives while suppressing others.

2.4 Algorithmic intensification of social comparison

While the previous section outlined the concept and dimensions of algorithmic media content awareness, the following section discusses how algorithmic curation may influence social comparison processes on social media.

Besides the interface level features typically present on social media, algorithmic curation shapes what users are exposed to in their feeds, which can amplify the effects of social comparison processes even more. As mentioned in Wenninger et al. (2021), engagement is a key factor in deciding on what is prioritized by Facebook's algorithm. Thus, a regular user is more likely to see content that already has a high number of likes and comments, making upward comparisons in terms of metric based digital status more likely. Algorithmic awareness could theoretically buffer against the negative effects of such social comparisons in a number of ways. First, feeling a sense of control over the ways in which one could attain the desired object or state can ameliorate envy. Knowledge regarding how algorithms work and how to optimize for algorithmic prioritization could offer users a sense of control over the process towards achieving high engagement and a higher metric based social status (Meythaler et al., 2025). Moreover, simply having an awareness of the fact that one is more likely to encounter content with high engagement when browsing social media could put any unfavorable comparisons of one's own engagement metrics into perspective.

For a passive user, meaning someone who simply browses social media platforms without actively posting, commenting or creating content (Turel, 2021), there are other factors at play. The positivity bias that is prevalent on SNS (Wenninger et al., 2021), combined with the fact that influencers often aim to portray themselves leading an aspirational lifestyle characterized by luxuries and material wealth (Meythaler et al., 2025), makes it more likely that a passive user is

exposed to these displays of envy-inducing status markers through algorithmic curation, regardless of whether one is actively searching for it or not. Superfluous, extravagant and out-of-the ordinary content is also likely to foster more engagement, a key factor in what is prioritized by the algorithm (Wenninger et al, 2021). Furthermore, the economic incentives guiding recommender systems (Carrasco-Farré et al., 2025), plus the observation that envy can trigger subjects to want to buy more (Krasnova et al., 2015; Wenninger et al., 2021), makes algorithmic prioritization of envy-inducing content more likely. Again, algorithmic awareness could theoretically buffer against negative effects of envy arising from algorithmic amplification of exposure to content that lends itself to upward comparisons, as one can engage more critically with the picture being painted of other people's lifestyles, as well as with personalized advertisements.

Additionally, recommendation and search algorithms facilitate the discovery of similar others through features such as friend suggestions and content filtering based on existing networks and interaction patterns (Krasnova et al., 2015). This increases the likelihood that users are exposed to comparison targets who are perceived as similar to themselves, which according to social comparison theory heightens the emotional impact of upward comparisons in self-relevant domains (Wenninger et al., 2021).

2.5 Hypothesis Development

Building on the theoretical perspectives outlined above, algorithmic media content awareness may play an important role in shaping social comparison processes on social media. Social comparison theory suggests that individuals evaluate themselves by comparing their own attributes and achievements to those of others, particularly in domains that are personally relevant (Festinger, 1954; Wenninger et al., 2021). Social networking sites amplify these processes by facilitating frequent exposure to positively curated self-presentations, idealized portrayals of others' lives, and visible engagement metrics that function as indicators of social status (Krasnova et al., 2015; Meythaler et al., 2025; Wenninger et al., 2021).

Research on social media literacy further suggests that critical awareness of how online content is constructed can mitigate some of the negative consequences associated with such comparisons. Recognizing that social media portrayals are often selectively curated, edited, or strategically presented may reduce the tendency to interpret them as realistic standards for self-evaluation (Paxton et al., 2022).

Algorithmic media content awareness extends this critical perspective by also drawing attention to the fact that content exposure on social media is selectively tailored to maximize user engagement and potential revenue from personalized advertisements (Carrasco-Farré et al., 2025). Thus, users may be systematically exposed to content that lends itself to upward

comparisons, and awareness of these mechanisms may help users contextualize the content they encounter and engage with it more critically. More specifically, users who are aware that social media feeds are shaped by engagement-driven algorithms may be less likely to interpret the content they encounter as representative of everyday reality or as an accurate standard for self-evaluation. Such awareness may also encourage more intentional social media use, for example by influencing what users choose to engage with or avoid interacting with. Consequently, the following hypotheses are proposed:

H0*. There is no statistically significant correlation between algorithmic media content awareness and negative social comparison among upper secondary school students.*

H1. *Algorithmic media content awareness is negatively related to negative social comparison among upper secondary school students.*

3. Method

This chapter describes how the study was carried out and explains the choices that were made throughout the research process.

First, the overall research design is presented, including why a quantitative, cross-sectional approach was chosen. Then, the participants and sampling method are presented. After that, I describe how algorithmic media content awareness and negative social comparison were measured and how the survey was constructed.

The chapter then explains how the data was collected in the school setting, followed by a description of how the data was analysed in order to test the research hypothesis. Finally, ethical considerations are discussed.

Together, these sections provide an overview of how the study was conducted and how the results were obtained.

3.1 Research Design

This study adopted a quantitative, cross-sectional survey design to examine the relationship between algorithmic media content awareness and negative social comparison among upper secondary school students. A quantitative design involves collecting numerical data in order to identify patterns and relationships between variables across a larger group (Venkatesh et al., 2013). This approach is particularly suitable when the aim is to examine associations rather than to gain in-depth insights into individual experiences (Anderson & Aydin, 2005).

I started by researching the literature related to social comparison, social media literacy, and algorithmic media content awareness. I primarily searched for articles using the university library database provided by Luleå University of Technology. To find relevant studies, I used combinations of keywords related to algorithmic awareness (e.g., “algorithmic awareness”, “algorithmic curation”, and “content recommendation”), social media (e.g., “social media”, “Instagram”, and “TikTok”), and social comparison (e.g., “social comparison”, “self-perception”, and “well-being”). I also identified additional sources through backward citation searching, meaning that I looked at citations from previously identified articles to find relevant material. From this theoretical background, I formulated expectations about a potential relationship between algorithmic media content awareness and negative social comparison. I then examined this potential relationship using empirical, quantitative data.

This way of going from existing theory to empirical testing is characteristic of a deductive approach (Jacobsen, 2017). In deductive research, previous theory and research are used as the

basis for formulating hypotheses that are later tested empirically. This differs from an inductive approach, where the researcher instead starts from empirical observations and uses these to generate new theoretical insights (Jacobsen, 2017). Inductive approaches are also more commonly associated with qualitative research aiming to gain a deeper understanding of participants' perspectives and lived experiences (Kaplan & Maxwell, 2005), which was not my aim with the present study.

In this sense, I utilized a hypothesis-testing approach. As described by Pugh and Molinaro (2016), hypothesis testing involves formulating a hypothesis based on existing research, collecting data from a relevant sample of the population, and then performing statistical tests on the data to test whether or not the hypothesis is supported.

I collected the data during a limited period of time using a survey distributed to upper secondary school students. Since each participant responded to the survey on only one occasion, I examined the variables as they appeared at that specific point in time. In other words, I captured a "snapshot" of the relationship between algorithmic media content awareness and negative social comparison during the period of data collection. According to Jacobsen (2017), this type of research design is referred to as a cross-sectional design. This differs from longitudinal studies, where the same participants are studied repeatedly over a longer period of time in order to examine changes and potential causal directions. It also differs from experimental designs, where researchers actively manipulate variables under controlled conditions in order to investigate causal relationships (Jacobsen, 2017).

While cross-sectional studies do not allow for conclusions about causality in the same way as longitudinal or experimental designs, they are well suited for identifying patterns of co-variation between clearly defined variables (Maier et al., 2023). Given that my aim was to examine whether a relationship exists between algorithmic media content awareness and negative social comparison, rather than to establish cause-and-effect relationships, I considered a cross-sectional design appropriate.

3.2 Participants and Sampling

I recruited participants from a local upper secondary school in a smaller town in central Sweden. Although the name of the school or exact location is not disclosed due to confidentiality reasons, it should be noted that all participants were enrolled in vocational programs, and included students from first, second, and third year.

In total, 63 students participated in the study (27 women and 36 men).

As my sampling method, I used convenience sampling. According to Stratton (2021), convenience sampling refers to when participants are recruited based on accessibility and willingness to participate, rather than through random selection procedures. In this study, participants were selected based on their availability in a school setting and their willingness to take part in the survey.

Non-probability sampling methods, such as convenience sampling, differ from probability sampling in that not all members of the target population have an equal chance of being selected. As a result, convenience sampling does not allow for generalization to a broader population, and findings are instead limited to the particular sample used in the study (Stratton, 2021). Ideally, if I wanted to generalize my findings to all upper secondary school students, I would have to include students from various geographical regions and cultural backgrounds, in different programs and educational settings, etc. No one category of student should be more or less likely to be included in the sample studied.

Despite these limitations of convenience sampling, Battaglia (2008) mentions that convenience sampling can still be of value for researchers when time and resources are limited, as long as these limitations are recognized. In the context of this study, this means that the observed results may not be generalizable to a wider population of students.

3.3 Measures and Operationalization

Operationalization refers to the process of translating abstract concepts into measurable variables (Martinez, 2017). Since constructs such as algorithmic awareness and social comparison cannot be observed directly, they must be represented through survey items that capture different aspects of the underlying phenomena.

For these purposes, I used and adapted scales and items developed in previous research (De Vries & Kühne, 2015; Zarouali et al., 2021). As discussed in Venkatesh et al. (2013), measurement validity refers to how well an instrument (such as a survey) is able to actually capture and quantify the phenomenon of interest. Thus, measurement validity is an important aspect when it comes to the legitimacy of quantitative research. By basing the survey used in this study on scales and items previously used in academic research, I increased the likelihood of obtaining an adequate level of measurement validity.

I operationalized algorithmic media content awareness as participants' perceived understanding of how social media platforms curate and prioritize content in their feeds. I measured this construct using items adapted from the AMCA scale developed by Zarouali et al. (2021). The AMCA-scale consists of 13 items in total, and the participant is asked to indicate to which extent

they are aware of the statements made in each item on a 5-point rating scale, ranging from “not at all aware” to “completely aware”.

The adaptations I made to the AMCA-scale to fit the context of this study include the following; First, I translated all items from English to Swedish, since I administered the survey in a Swedish school setting. I also excluded one item from the original scale. Item HAI2 (human-algorithm interplay, item 2) contains the following statement: “The [media content] that algorithms recommend to me on [platform name] depend on my online behavioral data” (Zarouali et al., 2021, s. 8). When I compared it to the previous item in the scale HA11 (human-algorithm interplay, item 1) “The [media content] that algorithms recommend to me on [platform name] depend on my online behavior on that platform” (Zarouali et al., 2021, s. 8), I considered the distinction between “online behavior on that platform” and “online behavioral data” too technical for the target group. To avoid confusion, I excluded HA12.

I operationalized negative social comparison as the degree to which one feels that others are doing better than oneself, stemming from comparisons made with content seen on social media. To measure this construct, I adapted items from De Vries and Kühne (2015). In their study on facebook usage and negative social comparisons, De Vries and Kühne (2015) used the following two statements to measure the degree of negative social comparison: “When I read news feeds (or see photos of others) I often think that 1) others have better lives than I do; 2) others are doing better than I am” (s. 219). Participants were asked to rate the degree to which they agreed with each statement on a 5-point rating scale, ranging from “totally disagree” to “totally agree”. I translated these items to Swedish and adjusted the wording to refer to social media content more broadly.

I computed composite variables by averaging item responses, where higher values indicated higher levels of algorithmic awareness and negative social comparison. A composite variable is a single numerical value created by combining multiple related items, and is commonly used to represent complex constructs in quantitative research (Song et al., 2013).

In addition to the main constructs described above, I asked participants to report their age and gender. As mentioned in Wenninger et al. (2021), gender has been shown to moderate the effects of social comparisons made on social media in previous research. Researchers such as Paxton et al. (2022) and Schaben et al. (2026) also discuss how gender could potentially influence the effects of social media literacy interventions. Thus, I included gender in the survey to allow for future analysis of potential moderation effects. Although I did not identify any prior studies that explicitly examined the relationship between age and algorithmic awareness or negative social comparison, I also included age as a control variable due to its theoretical relevance in relation to cognitive development and critical media literacy during adolescence.

To view the survey in its entirety, please refer to Appendix: Survey questions.

3.4 Data Collection Procedure

I collected data through an online survey administered to upper secondary school students. I created and distributed the survey using Google Forms, and participants accessed it by scanning a QR code provided in the classroom. I conducted the data collection during scheduled class time in March 2026.

A teacher at the school administered the survey in the classroom on my behalf, across multiple sessions involving different classes. The teacher provided the students with the QR code, and the students completed the survey individually using their own devices during class time.

Since participation was entirely voluntary and the survey was administered during scheduled class time, I could not determine an exact response rate. Although I could have asked the teacher to report the number of students present in each class so I could compare it to the number of submitted responses, I chose not to do so in order to avoid creating any perception of participation being mandatory. Instead, the teacher was instructed to emphasize that participation was voluntary and that students could choose not to participate without any consequences. The final sample therefore consists of those students who actively chose to take part in the study.

It should be noted that the study relied on self-reported survey data, meaning that participants themselves estimated their levels of algorithmic media content awareness and negative social comparison. Self-reported measures are often used in research because they provide a practical and cost-efficient way of collecting data from larger samples (Gran et al., 2021). However, such measures may also be influenced by factors such as self-perceptions and personality traits (for example, confidence), meaning that participants' responses may not always accurately reflect their actual level of awareness (Gran et al., 2021).

According to Anderson and Aydin (2005), survey research methodologies are typically used in information systems research when the intention is to explore quantitative relationships between variables. In contrast, qualitative methods are particularly useful when the aim is to gain an in-depth and nuanced understanding of participants' perspectives and lived experiences. Such rich understandings can for example be acquired through interviews centering around open-ended questions where the interviewee is encouraged to elaborate on their answers (Kaplan & Maxwell, 2005). Kaplan and Maxwell (2005) further argue that quantitative methods are well suited for exploring the potential existence of a relationship between variables, while qualitative methods are more capable of explaining how and why a relationship exists.

However, the aim of this study was not to explain underlying processes or individual experiences in depth, but rather to identify broader patterns among a larger group of students. While interviews could have provided richer and more detailed insights, they would also have required significantly more time and resources. In addition, interviews demand a greater level of commitment from participants, which may have reduced participation. A survey therefore enabled me to efficiently collect data from multiple participants within a limited time frame and to test the research hypothesis using statistical analyses, as described in the following section.

3.5 Data Analysis

I conducted the statistical analyses using JASP, an open-source statistics program (The JASP Team, 2025). Before conducting the analyses, I computed composite variables for the main variables. As described in a previous section, algorithmic media content awareness and negative social comparison were each measured using multiple survey items. To obtain overall measures of these constructs, I calculated the mean value of the respective items, resulting in two composite variables. Higher values indicated higher levels of each construct.

Descriptive statistics were calculated for both variables, including mean values, standard deviations, and minimum and maximum values. These statistics were also examined across subgroups based on gender (see Table 1) and age (see Table 2), in order to provide an overview of how the variables were distributed within the sample.

To test the hypothesis that algorithmic media content awareness is negatively related to negative social comparison, I conducted a Pearson correlation analysis. As described by Habibzadeh (2015), Pearson's correlation coefficient (r) is a statistical measure used to assess the strength and direction of a linear relationship between two variables. The coefficient ranges from -1 to +1, where values closer to -1 indicate a stronger negative relationship, values closer to +1 indicate a stronger positive relationship, and values around 0 indicate little to no linear relationship. Importantly, correlation does not imply causation, but rather indicates the extent to which two variables tend to vary together. Pearson's correlation is particularly suitable when analysing continuous variables (Habibzadeh, 2015), meaning variables that can take on a range of numerical values. As algorithmic media content awareness and negative social comparison were constructed as averaged composite variables, they were treated as continuous.

The analysis was conducted within the framework of hypothesis testing. Hypothesis testing is a method used to evaluate whether patterns observed in a sample are likely to reflect real relationships in a larger population, or whether they may have occurred by chance (Pugh & Molinaro, 2016). In this study, the aim was to test the hypothesis that algorithmic media content awareness is negatively related to negative social comparison among upper secondary school students. In order to assess this, the statistical significance of the correlation coefficient was

evaluated using a p-value. The p-value represents the probability of observing a result at least as extreme as the one obtained, assuming that no actual relationship exists in the population. Smaller p-values therefore indicate that the observed relationship is less likely to be due to random variation (Pugh & Molinaro, 2016).

In line with common practice in the social sciences, a significance level of 0.05 was used as the threshold for statistical significance (e.g. Salkind, 2007). This means that results with a p-value below 0.05 were considered unlikely to have occurred by chance. However, statistical significance should be interpreted with caution, as it does not indicate the size or practical importance of a relationship (e.g. Sen et al., 2022), but only whether it is likely to reflect a pattern beyond random variation.

In addition to the main analysis, exploratory correlation analyses were conducted between individual subdimensions of algorithmic media content awareness and individual items of negative social comparison (see Table 3). Exploratory analyses refer to analyses that are not specified in advance, but are instead conducted to identify potential patterns or relationships in the data (Andrade, 2023). Since these analyses were not part of the original hypothesis, they should be interpreted with caution. Conducting multiple statistical tests increases the risk of identifying statistically significant results by chance alone, a phenomenon known as Type I error (Pugh & Molinaro, 2016). This means that some observed relationships may appear significant even if no true relationship exists in the population.

At the same time, failing to detect a relationship that actually exists (a Type II error) is also possible, particularly in smaller samples (Pugh & Molinaro, 2016). Because of these limitations, the exploratory analyses are not used to draw definitive conclusions, but rather to identify potential patterns that may be of interest for further investigation. As such, they serve primarily as a basis for discussion and for suggesting directions for future research.

3.6 Ethical considerations

I took several steps to ensure that the study adhered to ethical guidelines. As stated by the Swedish Research Council (2024), informed consent is a key part of good research practice. I ensured informed consent by clearly informing participants that their participation in the study was voluntary, and that even if they were to start the survey, they could discontinue at any point without giving a reason why. This was communicated both by the teacher who was present during completion of the survey, as well as by a written statement in the survey before accessing survey questions.

Moreover, the “informed” part of informed consent highlights the need for participants to have some understanding of the research they are agreeing to partake in. To ensure that consent was

informed, I provided a written explanation of the study's purpose, how the data will be used, and that the survey is anonymous. I also briefly explained the research question while avoiding details that could influence responses. I discussed suitable phrasings with the supervisor before the survey was administered.

Additionally, the Swedish Research Council (2024) also highlights requirements associated with the collection of personal data. Personal data refers to data which can be tied to a particular person and used to identify said person, either alone or in combination with additional data sources (Swedish Research Council, 2024). As stated before, I did not collect any personal data that could identify participants. The survey was anonymous, and I only collected age and gender for descriptive purposes.

Lastly, because some participants were under the age of 18, I obtained permission from the school principal before conducting the study. I first contacted the principal to ask whether it was appropriate to carry out the study at the school. At the principal's request, I shared the survey for review. After reviewing the survey, the principal approved the study.

4. Results and Analysis

4.1 Descriptive statistics

I calculated descriptive statistics for the variables algorithmic media content awareness and negative social comparison. Table 1 presents the mean values, standard deviations, and observed minimum and maximum scores for these variables, grouped by gender. Table 2 presents the corresponding descriptive statistics across different age categories.

Both algorithmic media content awareness and negative social comparison were measured using 5-point rating scales, meaning that the possible values for the composite variables ranged from 1 to 5. Higher values indicate higher levels of algorithmic media content awareness and negative social comparison, respectively.

Overall, the results indicate moderate levels of both algorithmic media content awareness and negative social comparison in the sample. At the same time, some differences can be observed between groups. These differences are descriptive, and no statistical tests were conducted to examine whether they are statistically significant.

With regard to gender, male participants reported somewhat higher levels of algorithmic media content awareness (M = 3.69, SD = 1.01) compared to female participants (M = 3.32, SD = 0.98). In contrast, female participants reported higher levels of negative social comparison (M = 2.94, SD = 1.10) than male participants (M = 2.33, SD = 0.85).

Table 1

Descriptive statistics grouped by gender

	AMCA_mean		NSC_mean	
	Women	Men	Women	Men
Valid	27	36	27	36
Missing	0	0	0	0
Mean	3.318	3.690	2.944	2.333
Std. Deviation	0.997	1.012	1.104	0.845
Minimum	1.750	1.667	1.000	1.000
Maximum	5.000	5.000	5.000	5.000

Note: AMCA_mean, Algorithmic media content awareness mean; NSC_mean, Negative social comparison mean.

When examining the results across age groups, a similar overall pattern can be observed. The mean levels of algorithmic media content awareness were relatively similar for participants aged

16 to 18, ranging from 3.43 to 3.56. This suggests that there were no clear differences in awareness levels between these age groups in the sample.

For participants aged 19 and those who selected “other”, the mean values differed somewhat. However, these groups consisted of very few participants (n = 2 and n = 3, respectively), making it difficult to meaningfully interpret these values.

A similar pattern was observed for negative social comparison. The mean values for participants aged 16 to 18 ranged from 2.63 to 2.85, indicating relatively similar levels across these groups. Some variation can be observed in the smaller groups, particularly for the “other” category, where the mean value was notably lower. However, as with algorithmic media content awareness, these results should be interpreted with caution due to the small number of observations.

Table 2

Descriptive statistics grouped by age

	AMCA_mean					NSC_mean				
	16	17	18	19	Other	16	17	18	19	Other
Valid	28	20	10	2	3	28	20	10	2	3
Missing	0	0	0	0	0	0	0	0	0	0
Mean	3.426	3.558	3.550	3.167	4.500	2.696	2.625	2.850	2.000	1.000
Std. Deviation	1.011	1.003	0.991	1.886	0.795	1.012	0.686	1.355	0.000	0.000
Minimum	1.750	1.667	2.250	1.833	3.583	1.000	1.500	1.500	2.000	1.000
Maximum	5.000	5.000	5.000	4.500	5.000	5.000	4.000	5.000	2.000	1.000

Note: AMCA_mean, Algorithmic media content awareness mean; NSC_mean, Negative social comparison mean.

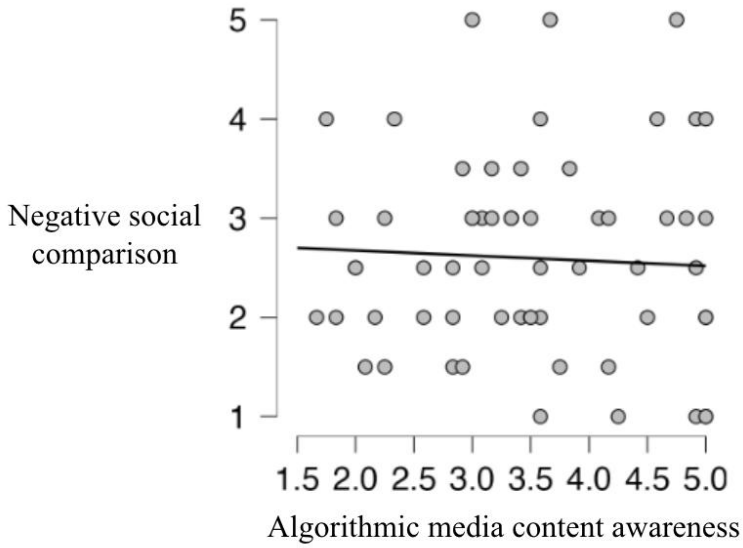
4.2 Correlation analysis

I conducted a Pearson correlation analysis to examine the relationship between algorithmic media content awareness and negative social comparison.

The results indicated a very weak negative correlation between the variables (r = -0.052), suggesting that higher levels of algorithmic media content awareness were associated with slightly lower levels of negative social comparison. However, the relationship was very small in magnitude. Figure 1 illustrates the relationship between the variables in a scatter plot.

Figure 1

Scatter plot illustrating the relationship between algorithmic media content awareness and negative social comparison.



The correlation was not statistically significant (p = 0.342), indicating that the observed relationship could likely be due to random variation in the sample rather than reflecting a systematic relationship in the population.

Taken together, these results provide no empirical support for the hypothesis that algorithmic media content awareness is negatively related to negative social comparison among upper secondary school students.

4.3 Exploratory analyses

In addition to the main analysis, I conducted exploratory correlation analyses between the subdimensions of algorithmic media content awareness and both the composite measure and individual items of negative social comparison. The results are presented in Table 3.

Table 3

Exploratory correlations

Variable		FIL	ADM	HAI	ETH
1. FIL	Pearson's r	—			
	p-value	—			
2. ADM	Pearson's r	0.864	—		
	p-value	1.000	—		
3. HAI	Pearson's r	0.770	0.794	—	
	p-value	1.000	1.000	—	
4. ETH	Pearson's r	0.803	0.879	0.753	—
	p-value	1.000	1.000	1.000	—
5. NSC_mean	Pearson's r	−0.061	−0.043	−0.125	0.014
	p-value	0.317	0.368	0.164	0.543
6. Others have better lives than I do	Pearson's r	0.038	0.061	0.035	0.119
	p-value	0.617	0.682	0.609	0.824
7. Others are doing better than I am	Pearson's r	−0.156	−0.146	−0.273*	−0.101
	p-value	0.111	0.126	0.015	0.215

Note. All tests one-tailed, for negative correlation.
* p < .05, ** p < .01, *** p < .001, one-tailed

FIL, Filtering; ADM, Automated decision-making; HAI, Human-algorithm interplay; ETH, Ethical considerations; NSC_mean, Negative social comparison mean.

When examining the correlations between the subdimensions and the composite variable for negative social comparison, all relationships were very weak (r ranging from -0.13 to 0.01) and non-significant. This indicates that there was no clear relationship between the different aspects of algorithmic awareness and overall levels of negative social comparison in the sample.

A similar pattern was observed for the item “others have better lives than I do”. All correlations with the subdimensions were close to zero (r ranging from 0.04 to 0.12) and non-significant, suggesting that this aspect of negative social comparison was not clearly related to any of the subdimensions of algorithmic media content awareness.

For the item “others are doing better than I am”, all correlations were negative, although still relatively weak. The strongest relationship was observed for the subdimension human-algorithm interplay ($r = -0.273$, $p = 0.015$), which was also statistically significant. This suggests that higher levels of perceived understanding of the interaction between users and algorithms were associated with lower agreement with the statement that others are doing better than oneself. The remaining subdimensions (filtering, automated decision-making, and ethical considerations) showed weak and non-significant correlations with this item (r ranging from -0.10 to -0.16).

While one statistically significant association was identified, this finding should be interpreted with caution given the number of tests conducted and the exploratory nature of the analyses. As discussed in the method section, conducting multiple statistical tests increases the likelihood of identifying significant results by chance alone.

5. Discussion

This study explored the correlation between algorithmic media content awareness and negative social comparison in a sample of upper secondary school students. More specifically, the study tested the hypothesis that algorithmic media content awareness is negatively related to negative social comparison among upper secondary school students. No significant correlation was found between the main constructs. Thus, H1 was not supported.

One possible explanation for the absence of a significant relationship is that algorithmic awareness alone may not be enough to influence social comparison processes on social media. As mentioned in Paxton et al. (2022) in the context of social media literacy interventions, combining different intervention approaches may strengthen the potency of each individual one.

Furthermore, Paxton et al. (2022) brings attention to the complexity characterizing the relationship between social media literacy skills and impacts on self-perception. The authors note that potential relationships could be non-linear, especially considering that the acquisition of social media literacy skills may be particularly sought out by individuals who are more vulnerable to negative effects of comparisons on social media and/or spend more time viewing appearance-related content. When considering how users tend to form their beliefs and understandings about platform algorithms through interactions with these recommender systems (Issar, 2024), and findings showing how standard social media features can amplify social comparison processes (e.g. Meythaler et al., 2025; Wenninger et al., 2021), it is possible that algorithmic media content awareness could also be positively related to negative social comparisons. Contrary to the hypothesis originally formed in the present study, if time spent on social media is positively linked to both self-reported measures of algorithmic awareness and negative social comparison, a positive association between these two constructs would emerge due to the presence of a confounding variable. Thus, it is theoretically possible to conceptualize both a negative and a positive association between algorithmic media content awareness and negative social comparison, through different (theoretical) mechanisms of action. These complexities related to potential cause and effect relationships could be explored more thoroughly with the use of longitudinal or experimental studies (Maier et al., 2023).

Another possible explanation relates to the broadness of algorithmic media content awareness as a construct. As explained previously, the conceptualization of algorithmic media content awareness by Zarouali et al. (2021) includes four separate subdimensions (filtering, automated decision-making, human-algorithm interplay, and ethical considerations). Since these subdimensions reflect different aspects of algorithmic awareness, they may also differ in how relevant they are for social comparison processes on social media. For example, while the subdimension “ethical considerations” can be considered highly relevant in the context of analyzing broader societal implications of algorithmic media curation, such as algorithmic bias reinforcing and amplifying intergroup hostility (Carrasco-Farré et al., 2025), being able to

critically reflect upon ethical concerns may not have the same relevance in terms of comparing and evaluating the self.

However, one subdimension that may potentially be especially relevant in the context of social comparison is human-algorithm interplay. As seen in the exploratory analyses of this study, this particular subdimension was negatively and significantly correlated with one of the items measuring negative social comparison (”others are doing better than I am”). While this finding ought to be interpreted with caution, given that it was part of the exploratory analyses, it points to a potentially interesting aspect of how algorithmic awareness could influence social comparison on a mechanistic level.

When viewed through a socio-technical lens, human-algorithm interplay stands out from the other subdimensions in the sense that it captures the essence of how human interaction with technical systems influences the behavior of these systems, which then has the potential to alter how the user interacts with the system, and so on, influencing each other in a continuous feedback-loop (Carrasco-Farré et al., 2025). Thus, awareness of this particular subdimension means recognizing how one’s own behavior on social media platforms can influence what one is subsequently exposed to, both in terms of recommended content and personalized advertisements (Newell & Marabelli, 2015). And, as mentioned in Paxton et al. (2022), one aspect of social media literacy concerns the intentionality with which one seeks out particular forms of content over others. By deliberately seeking out content that inspires positive feelings and avoiding content that centers around the pursuit of appearance ideals, some users actively limit their exposure to content that causes negative feelings to arise from negative social comparisons (Paxton et al., 2022). In today’s social media landscape, where users on platforms like Instagram, Facebook and TikTok spend the majority of their time viewing algorithmically recommended posts and videos (Federal Trade Commission v. Meta Platforms, Inc., 2025), having an awareness of how human-algorithm interplay functions is fundamental to being able to strategically alter one’s content exposure. For example, a student who is aware that liking and engaging with certain types of content will lead to more similar content being shown may actively try to avoid engaging with appearance-focused posts.

This interpretation is in line with public recommendations provided by the Public Health Agency of Sweden (2025a). On their website, they highlight how social comparisons on social media can be harmful for the well-being of young people and raises concerns about the impacts of algorithmically curated online environments. At the same time, they provide concrete advice on how social media users can influence what they see in their feed through intentional and strategic engagement with social media platforms. This becomes particularly relevant in light of recent discussions in the media regarding the potentially detrimental effects of algorithms on social media (Bergendorff, 2025), as it provides young people with tools needed to navigate an increasingly more complex and curated online environment. While the Public Health Agency of

Sweden (2025a) does not explicitly relate strategic engagement with algorithmically curated content to reduced negative social comparison, the exploratory findings of the present study suggest that such a relationship may be worth investigating further.

5.1 Limitations

This study has several limitations that should be acknowledged. First, the sample size was relatively small (N = 63), which means that weaker relationships may have been more difficult to detect statistically. As discussed previously in relation to Type II errors, smaller samples increase the risk that existing relationships in the population are not identified as statistically significant in the sample (Pugh & Molinaro, 2016). Thus, it is possible that a weak relationship between the variables may have gone undetected in the present study.

Second, the data was collected from students at a single upper secondary school in a smaller town in central Sweden, where all participants were enrolled in vocational programs. This limits the generalizability of the findings, as different patterns may have emerged in other educational or social contexts. For example, students enrolled in higher education preparatory programs, students from larger cities, or students from different socio-economic backgrounds may experience social media and algorithmic awareness differently. Third, the study relied on self-reported data, which as mentioned previously may be subject to various sources of error such as inaccuracies in self-perceptions. As discussed by Gran et al. (2021), participants' self-reported levels of awareness may be influenced by personality traits such as self-confidence.

Finally, another limitation concerns the level of analysis. This study focused on identifying general patterns between variables using quantitative data, rather than exploring students' personal experiences in depth. As a result, the study does not capture how students themselves interpret or make sense of algorithmic content or social comparison in their everyday lives. A more in-depth understanding of these processes would require qualitative approaches, such as interviews.

5.2 Future research

The findings from this study point to several directions for future research. First, it would be interesting to further investigate whether the individual subdimensions of algorithmic media content awareness differ in their relevance for social comparisons on social media. Rather than using a composite variable including all four subdimensions, future studies may benefit from analyzing the subdimensions separately. More specifically, the exploratory findings of this study suggests that human-algorithm interplay may be particularly relevant in the context of specific forms of social comparison, as compared to the other subdimensions.

Furthermore, the existence of a potentially non-linear or context-specific relationship between algorithmic media content awareness and negative social comparison warrants further investigation. Longitudinal or experimental designs could enable exploration of cause-and-effect mechanisms and shed more light on the presence of potential confounding factors, such as time spent on social media.

Studies with a more qualitative approach could also be beneficial in terms of investigating potential mechanisms and exploring individual experiences of algorithms and social comparisons on social media on a deeper level. Such approaches could provide more concrete insights into how these processes are experienced in everyday life through practical examples.

Finally, future studies should aim to include larger sample sizes and more diverse samples in order to enhance the generalizability of the findings across broader demographic groups.

5.3 Conclusion

In conclusion, this study did not find support for the hypothesis that algorithmic media content awareness is negatively related to negative social comparison among upper secondary school students. In other words, students with higher levels of algorithmic awareness did not appear to experience lower levels of negative social comparison on social media. However, the findings suggest that this relationship may be more complex than initially assumed. In particular, the findings indicate that certain subdimensions of algorithmic media content awareness could be more relevant than others, and that contextual factors may play an important part in influencing the relationship between algorithmic media content awareness and negative social comparison. By highlighting these complexities, the present study adds to the literature on algorithmic media content awareness and social comparison processes on social media while suggesting promising avenues for future research.

6. Final reflection

A broader ethical question raised by studies such as the present one concerns where responsibility should be placed when discussing the negative effects of algorithmically curated social media environments. Research on algorithmic awareness and social media literacy often focuses on how individuals can become more critical, reflective, and aware users. However, this may risk placing too much responsibility on users themselves to navigate highly complex technological systems designed to maximize engagement and attention.

At the same time, social media platforms are often characterized by limited transparency regarding how algorithmic recommendations are generated and prioritized (Bozdag, 2013; Shin et al., 2022). Because many recommendation systems function as "black boxes" (Newell & Marabelli, 2015), users may have limited opportunities to fully understand how their online environments are shaped. This raises questions about whether improving individual awareness alone is sufficient, or whether stronger regulatory frameworks and transparency requirements for platform companies are also necessary.

Another important consideration concerns digital inequality. Previous research by Gran et al. (2021) suggests that awareness of how algorithms work is unevenly distributed across the population. For example, younger and more highly educated individuals tend to report higher levels of algorithm awareness, while lower levels of awareness are more common among older individuals and those with lower levels of education. This raises important ethical questions regarding efforts to increase algorithmic awareness. If some groups are better equipped to understand and critically navigate algorithmically curated social media environments, they may also be better able to avoid harmful content and shape their online experiences in ways that benefit them. At the same time, other groups may have fewer opportunities or resources to develop the same level of understanding, which risks reinforcing existing social and digital inequalities.

These questions become particularly important in light of growing concerns regarding the psychological effects of social media algorithms on young people (e.g. Bergendorrf, 2025). While individual awareness and critical reflection are important, future discussions may also need to focus more broadly on the societal responsibilities of policymakers and technology companies in shaping healthier and more transparent digital environments.

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Appendix: Survey questions

Jag är…

- Man
- Kvinna
- Annat/Vill inte ange

Hur gammal är du?

- 16
- 17
- 18
- 19
- 20
- Annat

När jag tar del av innehåll på sociala medier tänker jag ofta att:

1. Original statement: Others have better lives than I do
   Swedish translation: Andra har bättre liv än jag
   1 Totally disagree/ Håller inte alls med
   2
   3
   4
   5 Totally agree/ Håller helt med

2. Original statement: Others are doing better than I am
   Swedish translation: Andra lyckas bättre än jag
   1 Totally disagree/ Håller inte alls med
   2
   3
   4
   5 Totally agree/ Håller helt med

Vänligen ange i vilken utsträckning du är medveten om följande påståenden om algoritmer i medieinnehåll:

1. Original statement: Algorithms are used to recommend [media content] to me on [platform name]
   Swedish translation: Algoritmer används för att rekommendera medieinnehåll till mig på sociala medier
   1 Not at all aware/ Inte alls medveten

2
3
4
5 Completely aware/ Helt medveten

2. Original statement: Algorithms are used to prioritize certain [media content] above others
   Swedish translation: Algoritmer används för att prioritera vissa typer av medieinnehåll framför andra
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

3. Original statement: Algorithms are used to tailor certain [media content] to me on [platform name]
   Swedish translation: Algoritmer används för att anpassa vissa typer av medieinnehåll specifikt till mig på sociala medier
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

4. Original statement: Algorithms are used to show someone else see different [media content] than I get to see on [platform name]
   Swedish translation: Algoritmer gör att andra personer kan få se annat medieinnehåll än det jag får se på sociala medier
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

5. Original statement: Algorithms are used to show me [media content] on [platform name] based on automated decisions
   Swedish translation: Algoritmer används för att visa mig medieinnehåll på sociala medier baserat på automatiserade beslut
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

6. Original statement: Algorithms do not require human judgments in deciding which [media content] to show me on [platform name]
   Swedish translation: Algoritmer kräver inte mänskliga bedömningar för att avgöra vilket medieinnehåll som visas för mig på sociala medier
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

7. Original statement: Algorithms make automated decisions on what [media content] I get to see on [platform name]
   Swedish translation: Algoritmer fattar automatiserade beslut om vilket medieinnehåll jag får se på sociala medier
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

8. Original statement: The [media content] that algorithms recommend to me on [platform name] depend on my online behavior on that platform
   Swedish translation: Det medieinnehåll som algoritmer rekommenderar till mig på sociala medier beror på hur jag beter mig på plattformen
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

9. Original statement: The [media content] that algorithms recommend to me on [platform name] depend on the data that I make available online
   Swedish translation: Det medieinnehåll som algoritmer rekommenderar till mig på sociala medier beror på den data som jag gör tillgänglig online
   1 Not at all aware/ Inte alls medveten
   2
   3
   4
   5 Completely aware/ Helt medveten

10. Original statement: It is not always transparent why algorithms decide to show me certain [media content] on [platform name]
    Swedish translation: Det är inte alltid tydligt varför algoritmer väljer att visa mig visst medieinnehåll på sociala medier
    1 Not at all aware/ Inte alls medveten

2
3
4
5 Completely aware/ Helt medveten

11. Original statement: The [media content] that algorithms recommend to me on [platform name] can be subjected to human biases such as prejudices and stereotypes
    Swedish translation: Det medieinnehåll som algoritmer rekommenderar till mig på sociala medier kan påverkas av mänsklig bias, såsom fördomar och stereotyper
    1 Not at all aware/ Inte alls medveten
    2
    3
    4
    5 Completely aware/ Helt medveten

12. Original statement: Algorithms use my personal data to recommend certain [media content] on [platform name], and this has consequences for my online privacy
    Swedish translation: Algoritmer använder mina personuppgifter för att rekommendera visst medieinnehåll på sociala medier, och detta kan få konsekvenser för min integritet online
    1 Not at all aware/ Inte alls medveten
    2
    3
    4
    5 Completely aware/ Helt medveten